Assignment 2 – Task 1: Titanic Passenger Survival Prediction System

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Deployment Link: https://ml-teaching-notebooks-jrrdwercxskw5kwc6rufvp.streamlit.app/

This notebook follows the complete Machine Learning lifecycle:

1. Data Collection
2. Data Understanding (EDA)
3. Data Pre-processing & Label Encoding
4. Model Training (SVC)
5. Model Evaluation
6. Making Predictions
7. Saving the Trained Model

Dataset: Titanic (built-in seaborn dataset)
Algorithm: Support Vector Classifier (SVC)

Step 0 – Import Libraries

In [1]:
```
import os, pickle, warnings
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
warnings.filterwarnings("ignore")
print("Libraries imported successfully OK")
```

```
Libraries imported successfully OK
```

Step 1 – Data Collection

Load the Titanic dataset (seaborn built-in).

In [2]:
```
df = sns.load_dataset("titanic")
print(f"Dataset Shape: {df.shape}")
df.head(10)
```

```
Dataset Shape: (891, 15)
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	
5	0	3	male	NaN	0	0	8.4583	Q	Third	man	
6	0	1	male	54.0	0	0	51.8625	S	First	man	
7	0	3	male	2.0	3	1	21.0750	S	Third	child	
8	1	3	female	27.0	0	2	11.1333	S	Third	woman	
9	1	2	female	14.0	1	0	30.0708	C	Second	child	

Step 2 – Data Understanding (EDA)

In [3]:
```
print("=== Dataset Info ===")
df.info()
```

```
=== Dataset Info ===
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    str
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    str
 8   class        891 non-null    category
 9   who          891 non-null    str
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    str
 13  alive        891 non-null    str
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 100.4 KB
```

In [4]:
```
print("=== Descriptive Statistics ===")
df.describe()
```

```
=== Descriptive Statistics ===
```

Out[4]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [5]:
```
print("=== Missing Values ===")
df.isnull().sum()
```

```
=== Missing Values ===
```

Out[5]:
```
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

In [6]:
```
print("=== Survival Distribution ===")
print(df["survived"].value_counts())

fig, axes = plt.subplots(2, 2, figsize=(12, 8))
fig.suptitle("Titanic – Exploratory Data Analysis", fontsize=14, fontweight="bold")

sns.countplot(data=df, x="survived", ax=axes[0,0], palette="Set2")
axes[0,0].set_title("Survival Count (0=No, 1=Yes)")

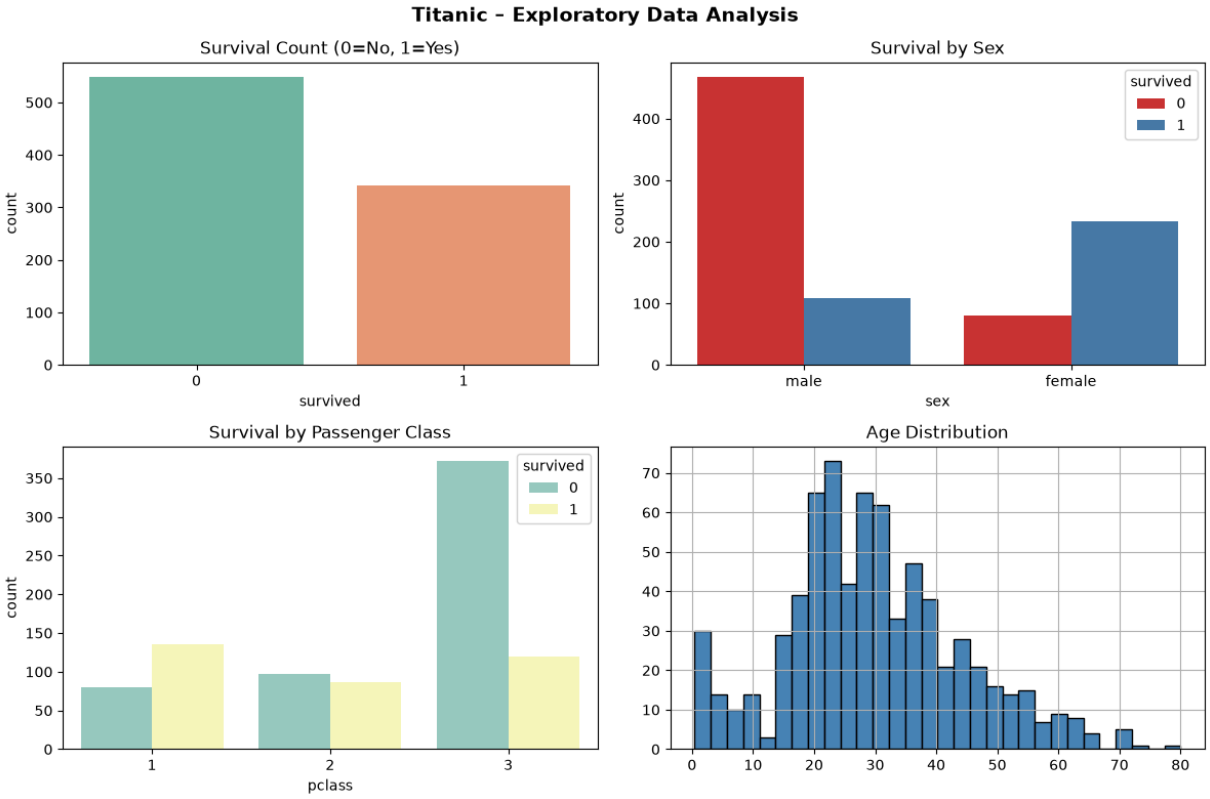
sns.countplot(data=df, x="sex", hue="survived", ax=axes[0,1], palette="Set1")
axes[0,1].set_title("Survival by Sex")

sns.countplot(data=df, x="pclass", hue="survived", ax=axes[1,0], palette="Set3")
axes[1,0].set_title("Survival by Passenger Class")

df["age"].dropna().hist(bins=30, ax=axes[1,1], color="steelblue", edgecolor="black"
axes[1,1].set_title("Age Distribution")
```

```
plt.tight_layout()
plt.savefig("titanic_eda.png", dpi=100)
plt.show()
print("EDA plots saved OK")
```

```
=== Survival Distribution ===
survived
0    549
1    342
Name: count, dtype: int64
```




```
EDA plots saved OK
```

Step 3 – Data Pre-processing & Label Encoding

In [7]:

```
# Select relevant features
features = ["pclass", "sex", "age", "sibsp", "parch", "fare", "embarked"]
target   = "survived"

df_model = df[features + [target]].copy()
df_model.dropna(inplace=True)
print(f"Shape after dropping nulls: {df_model.shape}")

# Label Encoding
le_sex      = LabelEncoder()
le_embarked = LabelEncoder()

df_model["sex"]      = le_sex.fit_transform(df_model["sex"])
df_model["embarked"] = le_embarked.fit_transform(df_model["embarked"])

# Save encoded dataset
```

```
df_model.to_csv("titanic_encoded.csv", index=False)
print("Encoded dataset saved to titanic_encoded.csv OK")
df_model.head()
```

```
Shape after dropping nulls: (712, 8)
Encoded dataset saved to titanic_encoded.csv OK
```

Out[7]:

	pclass	sex	age	sibsp	parch	fare	embarked	survived
0	3	1	22.0	1	0	7.2500	2	0
1	1	0	38.0	1	0	71.2833	0	1
2	3	0	26.0	0	0	7.9250	2	1
3	1	0	35.0	1	0	53.1000	2	1
4	3	1	35.0	0	0	8.0500	2	0

Step 4 – Model Training (Support Vector Classifier)

In [8]:

```
X = df_model[features]
y = df_model[target]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print(f"Training samples : {X_train.shape[0]}")
print(f"Testing  samples : {X_test.shape[0]}")

svc = SVC(kernel="rbf", random_state=42)
svc.fit(X_train, y_train)
print("\nSVC model trained successfully OK")
```

```
Training samples : 569
Testing  samples : 143

SVC model trained successfully OK
```

Step 5 – Model Evaluation

In [9]:

```
y_pred = svc.predict(X_test)
acc    = accuracy_score(y_test, y_pred)
cm     = confusion_matrix(y_test, y_pred)

print(f"Accuracy: {acc*100:.2f}%")
print("\nConfusion Matrix:")
print(cm)
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

fig, ax = plt.subplots(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["Did Not Survive","Survived"],
            yticklabels=["Did Not Survive","Survived"], ax=ax)
```

```
ax.set_xlabel("Predicted"); ax.set_ylabel("Actual")
ax.set_title("Titanic – Confusion Matrix")
plt.tight_layout()
plt.savefig("titanic_confusion_matrix.png", dpi=100)
plt.show()
```

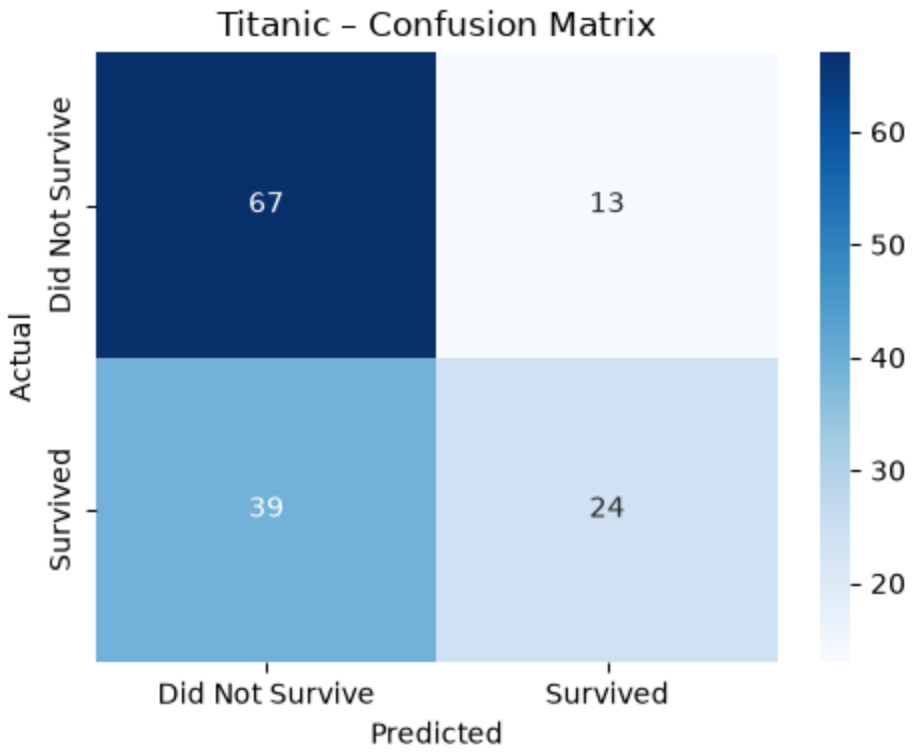
```
Accuracy: 63.64%

Confusion Matrix:
[[67 13]
 [39 24]]

Classification Report:
              precision    recall  f1-score   support

           0       0.63      0.84      0.72        80
           1       0.65      0.38      0.48        63

    accuracy                           0.64       143
   macro avg       0.64      0.61      0.60       143
weighted avg       0.64      0.64      0.61       143
```



Step 6 – Making Predictions on Sample Data

In [10]:
```
sample_raw = pd.DataFrame({
    "pclass":   [1,    3,    2],
    "sex":      ["female", "male", "female"],
    "age":      [29,   22,   35],
    "sibsp":    [0,    1,    1],
    "parch":    [0,    0,    0],
    "fare":     [211,  7.25, 26],
```

```
    "embarked": ["S",  "S",  "C"],
})
sample = sample_raw.copy()
sample["sex"]      = le_sex.transform(sample["sex"])
sample["embarked"] = le_embarked.transform(sample["embarked"])

predictions = svc.predict(sample)
sample_raw["Predicted_Survival"] = ["Survived ✓" if p==1 else "Did Not Survive ✗"
                                    for p in predictions]

# Save predictions
sample_raw.to_csv("titanic_predictions.csv", index=False)
print("Predictions saved to titanic_predictions.csv OK")
sample_raw
```

```
Predictions saved to titanic_predictions.csv OK
```

Out[10]:

	pclass	sex	age	sibsp	parch	fare	embarked	Predicted_Survival
0	1	female	29	0	0	211.00	S	Survived ✓
1	3	male	22	1	0	7.25	S	Did Not Survive ✗
2	2	female	35	1	0	26.00	C	Did Not Survive ✗

Step 7 – Save Trained Model

In [11]:

```
with open("titanic_svc_model.pkl", "wb") as f:
    pickle.dump(svc, f)

print("Model saved to titanic_svc_model.pkl OK")
print("\n" + "="*50)
print("TASK 1 COMPLETE")
print("="*50)
```

```
Model saved to titanic_svc_model.pkl OK

==================================================
TASK 1 COMPLETE
==================================================
```